

WEEK 1 PROGRESS REPORT

Multimodal AI E-commerce
Intelligence Automation Agent

Prepared by: Talha Nasir | Week: 1 | Status: ✓ Completed

7 Days	Backend	2 APIs	VLM
Duration	Status: Live	Modules Built	Integrated

Final Working Pipeline

Image Upload → VLM Analysis → Metadata Extraction → Embedding → Qdrant Search → Results

1. Project Introduction

This project builds a **Multimodal AI E-commerce Intelligence Automation Agent** that improves product discovery by combining text search and image search. Traditional keyword-based systems fail when users search with different words or upload images instead of typing. This system solves those limitations using semantic embeddings, vector databases, and Vision Language Models.

Core Technology Stack

FastAPI — Lightweight Python web framework with automatic Swagger docs

Qdrant — Vector database for semantic similarity search

Embeddings — Numerical vector representations enabling meaning-based search

VLM (qwen2.5vl:3b) — Vision Language Model for image understanding via Ollama

Docker — Container management for Qdrant service

GitHub — Version control and project management

2. Week 1 Story Summary

The week began with project architecture planning and FastAPI backend setup, followed by integrating Qdrant for vector search, adding the Vision Language Model for image analysis, connecting VLM output to product creation (analyze-and-save flow), and finally validating the complete multimodal pipeline with real product images.

***Key Achievement:** End-to-end image search confirmed with similarity scores above 0.90 for both brown shirt (0.904) and brown handbag (0.923) test cases.*

3. Day-by-Day Progress

Day 1 — Project Planning & Setup

Defined project architecture with modular separation of concerns. Selected FastAPI, Qdrant, VLM, Ollama, and GitHub as the core stack. Planned separate service files for products, embeddings, and VLM logic.

Key insight: Architecture is not just using a model — it requires proper API design, storage, and error handling.

Day 2 — FastAPI Backend Setup

Created the FastAPI project structure and initial product APIs (create, list, search). Configured Uvicorn server and verified Swagger documentation at <http://127.0.0.1:8000/docs>. Learned Python import path rules.

Key insight: FastAPI routes define how users communicate with the backend; each endpoint has a single responsibility.

Day 3 — Qdrant & Semantic Search

Integrated Qdrant vector database via Docker. Implemented embedding generation and stored product vectors in Qdrant. Enabled semantic search — 'sports shoes' can now match 'Nike Air Max Sneakers' by meaning.

Key insight: Vector databases enable meaning-based search over unstructured data, unlike traditional SQL.

Day 4 — Vision Language Model Integration

Added qwen2.5vl:3b through Ollama for product image understanding. VLM returns structured JSON with product_type, category, colors, style, material_guess, visible_features, search_tags, and description.

Key insight: VLM converts images to text metadata; that metadata then drives embedding and vector search.

Day 5 — Analyze-and-Save Flow

Connected VLM analysis with product creation. Uploading an image now automatically generates a product name, description, embedding, and Qdrant record — enabling semi-automated catalog creation from images.

Key insight: AI model output only becomes useful when connected to application logic, storage, and APIs.

Day 6 — Business Understanding

Analyzed the product from a SaaS perspective. Unlike general AI assistants, this system is specialized for e-commerce workflows: catalog ingestion, semantic search, and image-based discovery through clean APIs.

Key insight: A SaaS product becomes valuable by solving a specific business workflow, not just using an AI model.

Day 7 — Full Pipeline Testing & Optimization

Validated the complete multimodal pipeline with real images. Tested brown shirt (score 0.904) and brown handbag (score 0.923). Fixed import errors, Qdrant Docker issues, VLM slowness, and JSON parsing bugs.

Key insight: Building an AI system is mostly about connecting models with APIs, storage, search, and error handling.

4. Problems Faced & Solutions

#	Problem	Root Cause	Fix
1	Import Path Error	ModuleNotFoundError: No module named 'Backend'	Backend in 'app' the /backend directory so Python resolves 'from app
2	Qdrant Container Management	Qdrant not running before vector search	Use 'docker start qdrant' and verify with 'docker ps' before starting the
3	VLM Slow (CPU-only)	Image analysis took 3 min 52 sec initially	Compressed images to 512px max, used num_predict and num_ctx li
4	Missing Pillow Dependency	ModuleNotFoundError: No module named 'pillow'	pip install pillow — note: package name is 'pillow' but import is 'from P
5	VLM Scene Contamination	VLM described chair, phone, background	Instead of prompt explicitly tell the model to ignore background, pose,
6	Dict-in-List VLM Output	TypeError: sequence item 0: expected str	Added safe_join() and safe_text() helpers that normalize strings, lists,

5. System Architecture

Text Search Flow

User Text Query → Generate Embedding → Qdrant Vector Search → Return Similar Products

Image Search Flow

User Uploads Image
→ VLM Analyzes Image (product type, colors, style, tags)
→ Build Search Text from Metadata
→ Generate Embedding
→ Qdrant Vector Search
→ Return Similar Products

Analyze-and-Save Flow

User Uploads Product Image
→ VLM Extracts: type, category, colors, style, material, features, tags
→ Backend Generates Product Name & Description
→ Generate Embedding → Store in Qdrant

6. API Endpoints

Method	Endpoint	Description
GET	/api/v1/products/	List all products
POST	/api/v1/products/	Create a product manually
POST	/api/v1/products/search	Text-based product search
POST	/api/v1/products/analyze-and-save	Upload image → auto-create product
GET	/api/v1/multimodal/search/text	Semantic text search endpoint

POST	/api/v1/multimodal/search/image	Image-based similarity search
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7. Test Results

Test Image	VLM Detection	Score	Result
Brown Shirt	Male Brown Slim Fit Formal Shirt	0.9048	✓ PASS
Brown Handbag	Unisex Brown Saddle Bag Handbag	0.9237	✓ PASS

8. Week 1 Status Summary

Feature	Status
FastAPI Backend	Working
Qdrant Vector DB	Working
Text Semantic Search	Working
Image VLM Analysis	Working
Image Search Pipeline	Working
Analyze-and-Save Flow	Working
VLM Optimization	Improved
GitHub Version Control	Completed
Frontend (React)	Planned Week 2
Authentication	Planned Week 2
Cloud Deployment	Planned Future

9. Week 2 Roadmap

- Build React frontend with product cards
- Connect frontend to FastAPI APIs
- Add text search UI with live results
- Add image upload UI with drag-and-drop
- Display VLM analysis output to users
- Add loading state for slow image analysis
- Handle duplicate products in the database
- Improve result filtering and ranking
- Prepare complete end-to-end demo flow

This report documents Week 1 of the Multimodal AI E-commerce Intelligence Automation Agent project. The backend foundation is complete and validated. Week 2 will focus on frontend development and a user-facing demo.